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*A UNIFORM COHOMOLOGY THEORY FOR ALGEBRAS**

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The deformation theory for algebras[1] suggests that given any category $\mathfrak{C}$ of equationally defined algebras and homomorphisms (e.g., associative algebras, Lie algebras, Jordan algebras, commutative associative algebras, nilpotent algebras with fixed finite index), there exists a natural cohomology theory appropriate to $\mathfrak{C}$. This paper gives a definition of a cohomology theory for all such $\mathfrak{C}$ in a uniform way. However, only the groups H^n for $n \geq 2$ are defined, H^2 being introduced as a group of "singular" extensions. Fixing a suitable concept of extension normalizes H^2 and therewith the theory. (The coefficient ring need not be a field, nor extensions additively split.) The axioms[2] for a cohomology theory are satisfied, beginning with H^2, including the vanishing of H^n for $n > 2$ when the coefficient module is injective. It follows that in the presence of sufficiently many injectives, the theory is unique, i.e., any other satisfying the axioms and with the same H^2 must coincide with the present one.

1. *Definition of the Cohomology Groups.*—Given an algebra A in the category $\mathfrak{C}$, by a two-sided module P over A is meant a module over the coefficient ring together with linear maps $A \otimes P \to P$ and $P \otimes A \to P$ (the images under which of $a \otimes p$ and $p \otimes a$ will be denoted simply by ap and pa, respectively) such that the module direct sum $A + P$ with multiplication given by $(a_1, p_1)\ (a_2, p_2) = (a_1a_2, a_1p_2 + a_2p_1)$ is again an algebra of the category $\mathfrak{C}$. A *singular extension* $\mathbf{E}$ over A with kernel C is a short exact sequence $0 \to C \to B \to A \to 0$ with B in $\mathfrak{C}$ and $C^2 = 0$. It follows that C is in a natural way a two-sided module over A. An *n-fold extension* over A with initial term a two-sided A module P is, for $n = 2$, just a singular ex-

tension $O \to P \to B \to A \to O$, and for $n > 2$, an exact sequence $O \to P \to P_{n-1} \to \ldots \to P_2 \to C \to O$ of two-sided A modules and module homomorphisms together with a singular extension $O \to C \to B \to A \to O$; these may be combined into the exact sequence $O \to P \to P_{n-1} \to \ldots \to P_2 \to B \to A \to O$. Equivalence of sequences of the latter kind is definable exactly as in the Yoneda theory, and the classes can similarly be made into a group.

Definition: The nth cohomology group $H^n(A, P)$ of A with coefficients in a two-sided A-module P is, for $n \geq 2$, the group of equivalence classes of n-fold extensions over A with initial term P.

2. *Comparison with Various Known Theories.*—No explicit mention has been made in the definition of the coefficient ring K or of whether the theory is relative or absolute. Suppose first K is a field, in which case all extensions are split as K modules. When $\mathfrak{C}$ is the category of associative or Lie algebras, then the cohomology groups as defined here agree with the classical Hochschild and Chevalley-Eilenberg groups, respectively. This follows immediately from the fact that both satisfy the axioms and vanish on injectives. (In the Lie case, an "injective module" Q over a Lie algebra A is to be interpreted as an injective left module over the enveloping algebra A^e, and must in general be infinite dimensional. To show that $H^n(A, Q) = 0$, it is convenient to use the Cartan-Eilenberg reformulation of the Lie cohomology theory in terms of the Ext functor.[3])

A cohomology theory for the category of commutative associative algebras over a field has been proposed by Harrison. In characteristic 2 Harrison's theory differs from the present one, and it is conjectured that they are distinct for all finite characteristics but zero. For example, if A is a zero algebra of dimension 1 over a field of characteristic 2, and P an A module, then in the present theory $H^2(A, P) = P$ and $H^n(A, P) = 0$ for $n > 2$, while in the Harrison theory $H^2(A, P) = P$ and $H^n(A, P) = 0$ for $n > 2$ *except* for n a multiple of 4, but $H^{4m}(A, P) = P$. In particular, in the Harrison theory $H^{4m}(A, Q) \neq 0$ for an injective Q unless $Q = 0$. This anomalous result of the Harrison theory is due to the fact that the conditions imposed by him on a commutative cochain are the linearized forms of certain identities which, in characteristic 2 (and probably in all nonzero characteristics), are not recoverable by identifying distinct variables. It is conjectured that the theories agree in characteristic zero, this being equivalent to the conjecture that in characteristic zero the Harrison cohomology of an algebra with coefficients in an injective module vanishes. Fundamental results which should be expected of a commutative cohomology theory (concerning, in particular, tensor products and localization), which were proved by Harrison for his H^2 and H^3 only, have been shown by Barr in a forthcoming paper to hold here for all dimensions.

When K is not a field, then the absolute cohomology theory in the associative case is due to Mac Lane (for K the integers) and Shukla (for general K), and in the Lie case to Dixmier. These theories will agree with the present one if and only if they have the property that the cohomology vanishes when the coefficient module is injective. It is conjectured that this is the case.

A cohomology theory for groups is obtainable in the present context either by referring to the group ring, or better, by observing that the concepts of module and singular extension over a group (the latter here having the meaning of extension with abelian kernel) are defined. The concepts needed for the definition are there-

fore available, and it is again trivial from the axioms that the present cohomology theory agrees with the classical one of Eilenberg-Mac Lane. With this point of view it is clear that the present formulation of the cohomology theory is applicable in many categories, such as categories of sheaves, with suitable definitions of module and singular extension.

3. *Projective Resolutions.*—Fixing an algebra A in the equationally defined category $\mathfrak{C}$, a *morphism* of a singular extension $\mathbf{E}$: $O \to C \to B \to A \to O$ into another $\mathbf{E}'$: $O \to C' \to B' \to A \to O$ is, as usual, a pair of maps $C \to C'$, $B \to B'$ such that together with the identity map $A \to A$ the obvious diagram commutes. Given $\mathbf{E}$, an A module C', and an A module homomorphism $C \to C'$, there exists an extension $\mathbf{E}'$ unique up to equivalence and a corresponding morphism $\mathbf{E} \to \mathbf{E}'$. A singular extension $\mathbf{G}$: $O \to Y \to X \to A \to O$ is *generic*[4] if given any singular extension E of A there exists a morphism $\mathbf{G} \to \mathbf{E}$. Generic singular extensions exist, e.g., let A be represented as the quotient of a free algebra $\bar{X}$ by an ideal $\bar{Y}$. Then $\bar{Y}^2$ being an ideal of $\bar{X}$, we may set $Y = \bar{Y}/\bar{Y}^2$ and $X = \bar{X}/\bar{Y}^2$. Let morphisms γ_1, γ_2: $G \to E$ be given, with algebra homomorphisms f_1, f_2: $X \to B$ and A module homomorphisms ψ_1, ψ_2: $Y \to C$. Set $\varphi = f_1 - f_2$, a linear mapping of X into C, and observe that any A module is also an $\bar{X}$ module by virtue of the homomorphism $X \to A$. Then for x_1, x_2 in X one has $\varphi(x_1x_2) = f_1(x_1)\varphi(x_2) + \varphi(x_1)f_2(x_2) = x_1\varphi(x_2) + \varphi(x_1)x_2$, i.e., φ is a derivation of X into C. Denoting the group of these by $Z^1(X, C)$ and the map $\bar{Y} \to X$ by i, the group of equivalence classes of extensions over A with kernel the two-sided A module C is naturally isomorphic to Hom $(Y, C)/i^*Z^1(X, C)$. This, therefore, is naturally isomorphic to $H^2(Y, C)$.

Let $\xrightarrow{\partial} X_n \xrightarrow{\partial} X_{n-1} \xrightarrow{\partial} \cdots \xrightarrow{\partial} X_2 \to Y \to O$ be a projective resolution of Y as a 2-sided A module, and denote the composite map $X_2 \to Y \to X$ by $\bar{\partial}$. We call $\cdots \xrightarrow{\partial} X_n \xrightarrow{\partial} \cdots \xrightarrow{\partial} X_2 \xrightarrow{\bar{\partial}} X \to A \to O$ a *generalized projective resolution* of A. Given a two-sided A module C, we obtain the complex $\cdots \xleftarrow{\delta_n}$ Hom $(X_n, C) \xleftarrow{\delta_{n-1}} \cdots \xleftarrow{\delta_2}$ Hom $(X_2, C) \xleftarrow{\bar{\delta}} Z^1(X, C)$. We then have the following

THEOREM. *(i)* $H^2(A, C) = ker\ \delta_2/im\ \delta$, *(ii)* $H^n(A, C) = ker\ \delta_n/im\ \delta_{n-1}$ *for* $n > 2$.

The theorem states in effect that the cohomology of A with coefficients in C may be obtained from a projective resolution of the kernel of a generic singular extension. The same theorem is valid when A is a group, part (ii) being then the essential conclusion of the "cup product reduction theorem."[5] In the presence of sufficiently many projectives the foregoing theorem may be taken as definition of the cohomology. We have given precedence to the Yoneda approach since many important categories, such as categories of sheaves, do not admit sufficiently many projectives.

4. *Generalizations.*—Suppose given for a category of algebras containing an algebra A, a cohomology theory satisfying the exactness axiom, and suppose there are sufficiently many injectives Q in the category of A modules and homomorphisms, but that $H^n(A, Q)$ in general does not vanish. Set for every A module P and all m, $H^{m,0}(A, P) = H^m(A, P)$. Fixing a value of n, suppose there has been defined for some m and all pairs (r, s) of nonnegative integers with $n + r + s < m$ and all P a subgroup $H^{n+r,s}(A, P)$ of $H^{n+r+s}(A, P)$. Let $O \to P \to Q \to P' \to O$ be an exact

sequence of A modules in which Q is injective, let $n + r + s = m - 1$, and define $H^{n+r,s+1}(A, P)$ to be the image of $H^{n+r,s}(A, P)$ under the connecting homomorphism $H^{m-1}(A, P') \rightarrow H^m(A, P)$; this does not depend on the choice of the sequence. One then has a natural filtration $H^m(A, P) = H^{m,0}(A, P) \supset H^{m-1,1}(A, P) \supset \ldots \supset H^{n,m-n}(A, P)$. Defining now for $i > 0$ the groups $\bar{H}^{n+1}(A, P) = H^{n,i}(A, P)$, one obtains a new cohomology theory for A in which only the nth and higher cohomology groups are defined, but in which for $m > n$, $\bar{H}^m(A, Q) = 0$ for Q injective. The choice of n normalizes the theory. In the foregoing we have in effect taken $n = 2$.

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[1] Gerstenhaber, M., "On the deformation of rings and algebras," *Ann. Math.*, **79**, 59–103 (1964). The theory parallels closely the Froelicher-Kodaira-Nijenhuis-Spencer theory for manifolds.

[2] Cf. Mac Lane, S., *Homology* (New York: Academic Press, 1963), pp. 99, 117, 287. All author references not specifically cited here may be found in the extensive bibliography to this volume.

[3] As given in their *Homological Algebra* (Princeton University Press, 1956), p. 270. The convenience of this approach was pointed out to the author by Barr.

[4] The concept is implicit in Hochschild, G. P., "On the cohomology theory for associative algebras," *Ann. Math.*, **47**, 568–579 (1946).

[5] Cf. Mac Lane, *op. cit.*, p. 248.

QUANTUM CELL MODEL FOR HARD SPHERES

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Recently, there has been interest in the predictions of the cell model for quantum fluids. Results for an approximate cell model for liquid hydrogen have been published in these PROCEEDINGS.[1] In those investigations the cell potential was replaced by an approximate potential for which the energy levels could be easily evaluated. More recently, Henderson and Reed[2] have applied the cell model to liquid hydrogen using the correct cell potential.

Hillier and Walkley[3] have recently used the cell model to calculate the equation of state of a quantum system of hard spheres. However, they reported results only for molecules having particular diameters and masses. In this paper we have calculated all the thermodynamic properties of this interesting system. Further, we have shown that it is possible to define a characteristic energy which enables us to present data which is applicable to any quantum system of hard spheres. Hillier and Walkley[4] have suggested that the cell model overestimates quantum corrections. To investigate this suggestion we have applied the more reliable tunnel model[5] to a system of rigid spheres at absolute zero and have found that, at least at very low temperatures, the tunnel model shows greater quantum corrections than does the cell model.

Classical Hard Spheres.—The partition function for a system of classical hard spheres is given by[6]